

Creating 3D Personal Avatars with High Quality Facial Expressions for Telecommunication and Telepresence

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ABSTRACT

This study aims at providing a low-cost solution for telepresence where people are reconstructed as 3D avatars using an ordinary webcam, while still exhibiting abundant facial information (such as micro-expressions) that are critical for face-to-face communication. We estimate the basic 3D shape and texture of the body from a set of video frames, and then subsequently update its body pose, facial expression, and facial texture in each frame. Our method is expected to reduce the entry barrier of VR systems and create an embodied telecommunication that conveys rich information and subtle emotional changes to deepen mutual understanding at a distance.

Index Terms: Human-centered computing—Human computer interaction (HCI)—interactive systems and tools—User interface programming; Human-centered computing—Visualization—Visualization systems and tools

1 INTRODUCTION

Immersive virtual reality (VR) systems have widely been used in games, training, and simulations, but the dominant technology used for remote meetings is still videoconferencing that relies on video cameras and 2D screens. Although many users have reported fatigue and lack of unity using videoconference software [4], the requirement and cost of various devices necessary for current embodied VR systems prevents their general adoption by the society. Furthermore, compared with 2D videoconferencing, current metaverse systems using animated cartoon-like 3D avatars [1] cannot convey the detailed facial expressions that are important for deep communication. In fact, in daily face-to-face communication, subtle changes in the eyes, eyebrows, mouth, cheeks, forehead, and chin, which are often called micro-expressions, unconsciously express the state of mind of the person, and people use this information to read the personality of the other person and his/her emotions at that moment.

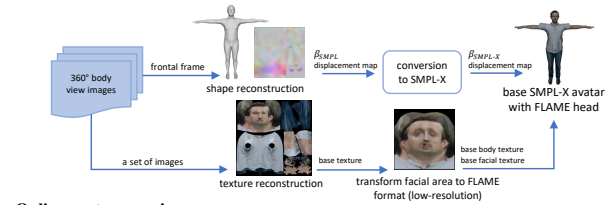
In this paper, we propose a method for creating 3D personal avatars with high quality facial expressions from a single ordinary RGB camera without the need for special sensors, markers, or a large number of cameras. Such 3D avatars can then be relocated, manipulated and modified to accommodate different types of viewing devices and different virtual environments. We implement the system on top of recent advancements in machine learning techniques for markerless body pose and facial expression tracking at real time speeds, as well as to produce body reconstruction from a set of 2D images. This method is expected to enable deep mutual understanding among remote people in a 3D shared environment, similar to face-to-face communication in real space.

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Offline avatar creation



Online avatar reposing and texturing

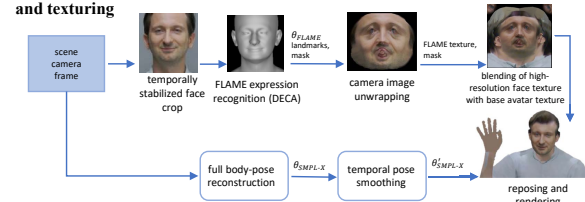


Figure 1: Overview of the system's processing pipeline.

2 OVERVIEW

The system consists of two parts (Fig. 1): an offline part that creates a base 3D avatar from a series of photos and a real-time part, where the avatar is reposed based on the tracking of user's body. In order to provide photo-realistic face representation, we combine body pose and facial expression recognition with the real-time texture data fed from the camera and overlay it on the reposed head model.

3 IMPLEMENTATION

Personalized avatars are created by registering user's body shape and texture. This is currently performed offline and takes a few minutes. During the online operation of the system, the base avatar is reposed using motion tracking data while the up-to-date facial texture and expressions are used to repose and render the face. The SMPL-X body model [8] and the FLAME head model [6] are used as the underlying representations of user's body.

3.1 Base avatar

Our registration process is based on the combination of the methods introduced by Alldieck et al. in [2] and [3], owing to their good quality and pretrained models available with libraries for inference. First, we record a short sequence of a user rotating in front of the camera. Then, we use a single frontal frame and the Tex2Shape [3] method to estimate overall body shape. We chose this method as it produces near instant results. Also, conveniently, it represents body shape as a vector β_{SMPL} of SMPL [7] body shape parameters and a displacement layer represented as a 2D map. The displacement layer is a simple and efficient way to model hair and clothes on top of the SMPL body shape. Since the output of the method is in the form of SMPL parameters, we obtain β_{SMPL-X} by finding a closely matching SMPL-X body shape via gradient descent.

In order to represent fine displacements, the Tex2Shape method relies on a modified, high-resolution version of the SMPL model (with 4 times the number of vertices). Thus, we have up-scaled the SMPL-X model in a similar manner and use its high resolution version throughout our processing pipeline.

The SMPL texture for the model is generated from a set of around 10 frames capturing the subject at different angles using the method introduced in [2]. Conveniently, the SMPL-X UV mapping is nearly identical to that of SMPL allowing to use the texture directly with SMPL-X avatars.

3.2 Motion tracking

In order to track users' body movements without markers, we employ the FrankMocap network [9], as it directly predicts body pose in SMPL-X format and operates at real-time speeds. The pose information θ_{SMPL-X} can then be used to repose the avatar. However, just like most of the currently published deep-networks, it operates on a frame-by-frame basis and this approach is prone to noisy predictions that results in unnatural, jittery movements when animated. Currently, we alleviate this problem by applying smoothing filter to the angles of each joint over a set of past frames.

3.3 Facial expressions

As the realistic facial representation was one of our primary goals, we use the highly expressive FLAME head model [6]. Recent advancements in deep-networks, such as DECA [5], allow to predict FLAME expression coefficients from a single facial frame at real-time speeds. However, while producing impressive results, we found that 3D heads with DECA-generated textures still fall short of fully conveying the original facial expression. This is partially due to the errors in expression recognition, but also due to the use of synthetic facial texture. Thus, in our approach, we decided to only rely on the FLAME expression vector θ_{FLAME} predicted by DECA and its dense landmark prediction. We use the latter to unwrap high resolution facial camera input into facial texture in the FLAME format. A live texture, coupled with high quality expression recognition results in a photo-realistic face that captures the same amount of detail as a video stream, yet is capable of being rendered and manipulated as a 3D model.

To alleviate the problems of jittery facial movements and skin texture "floating" around the face, we implemented a stabilized cropping frame around detected face. It adjusts its position between subsequent frames in smooth, incremental movements and avoids size adjustments unless necessary.

To complete the avatar, the areas of the face that were captured in a given frame are used to create a blending mask. We smoothen the edges of the mask in order to reduce visual discontinuities due to different level of detail, misalignments and differences in coloring, and use it to combine the high resolution facial texture with the low resolution base facial texture.

4 CONCLUSIONS

We created a prototype of low-cost telepresence system that employs live camera feed to produce 3D avatars with photo-realistic facial details. Many issues remain to be resolved in order to improve the tracking and reconstruction quality. Importantly, we observed that the current setup does not work well on persons with more complex hairstyles and clothes, and thus typically fares better on male subjects. We also observed deficiencies in the quality of both body and facial reconstruction that seem to be specific to Asian body types, likely due to their poor representation in the data sets that were used to train the deep networks employed by the system. We plan to address this in the future by training our own models on more balanced data sets. Behavioral and brain imaging experiments are also planned to study how different levels of expressiveness in

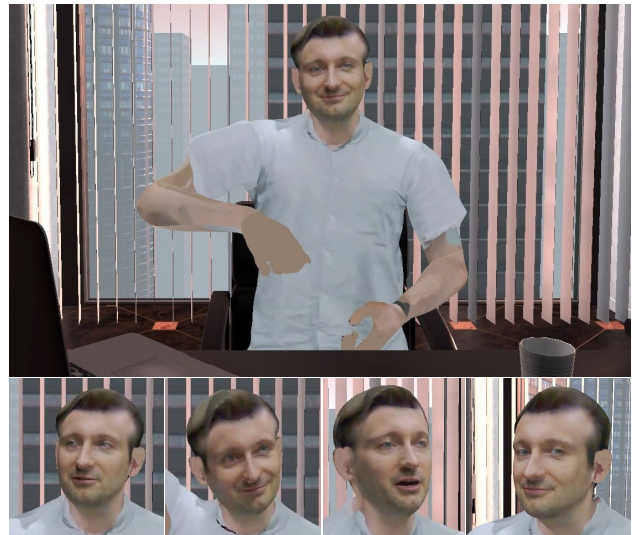


Figure 2: A 3D personal avatar gesturing in a virtual environment, with real-time facial details (top). Examples of captured facial expressions, including side views (bottom). See <https://youtu.be/P6M3F5L1ApA> for a short animated sequence.

avatars, as well as avatar manipulations, affect users' experience and affinity.

REFERENCES

- [1] A. Abdullah, J. Kolkmeier, V. Lo, and M. Neff. Videoconference and Embodied VR: Communication Patterns Across Task and Medium. *Proc. ACM Hum.-Comput. Interact.*, 5(CSCW2), oct 2021. doi: 10.1145/3479597
- [2] T. Alldieck, M. Magnor, W. Xu, C. Theobalt, and G. Pons-Moll. Detailed Human Avatars from Monocular Video. In *2018 International Conference on 3D Vision (3DV)*, pp. 98–109, Sept. 2018. ISSN: 2475-7888. doi: 10.1109/3DV.2018.00022
- [3] T. Alldieck, G. Pons-Moll, C. Theobalt, and M. Magnor. Tex2Shape: Detailed Full Human Body Geometry From a Single Image. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 2293–2303, Oct. 2019. ISSN: 2380-7504. doi: 10.1109/ICCV.2019.00238
- [4] J. N. Bailenson. Nonverbal Overload: A Theoretical Argument for the Causes of Zoom Fatigue. *Technology, Mind, and Behavior*, 2(1), 2 2021. <https://tmb.apaopen.org/pub/nonverbal-overload>. doi: 10.1037/tmb0000030
- [5] Y. Feng, H. Feng, M. J. Black, and T. Bolkart. Learning an animatable detailed 3D face model from in-the-wild images. *ACM Transactions on Graphics*, 40(4):88:1–88:13, July 2021. doi: 10.1145/3450626.3459936
- [6] T. Li, T. Bolkart, M. J. Black, H. Li, and J. Romero. Learning a model of facial shape and expression from 4D scans. *ACM Transactions on Graphics*, 36(6):194:1–194:17, Nov. 2017. doi: 10.1145/3130800.3130813
- [7] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black. SMPL: a skinned multi-person linear model. *ACM Transactions on Graphics*, 34(6):248:1–248:16, Oct. 2015. doi: 10.1145/2816795.2818013
- [8] G. Pavlakos, V. Choutas, N. Ghorbani, T. Bolkart, A. A. Osman, D. Tzionas, and M. J. Black. Expressive Body Capture: 3D Hands, Face, and Body from a Single Image. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10967–10977. IEEE Computer Society, Los Alamitos, CA, USA, jun 2019. doi: 10.1109/CVPR.2019.01123
- [9] Y. Rong, T. Shiratori, and H. Joo. FrankMocap: Fast Monocular 3D Hand and Body Motion Capture by Regression and Integration. In *IEEE International Conference on Computer Vision Workshops*, 2021.